Leela Gopala Bharath Kumar Challagulla

Full Stack Software Engineer

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SUMMARY

Full Stack Software Engineer with about 5+ years of experience building scalable, cloud-native web applications across healthcare, automotive, and public sector domains using React, Angular, TypeScript, Java Spring Boot, and Node.js in microservices and micro-frontend architectures. Experienced in designing GraphQL and REST APIs, event-driven systems with Kafka, and secure services using OAuth2/JWT and RBAC. Strong background in SQL/NoSQL databases, performance optimization, and cloud deployments on AWS and Azure with Kubernetes and Helm-based configurations. Proficient in CI/CD automation using GitHub Actions and Jenkins, Python scripting, and observability tooling, with a consistent focus on delivering reliable, high-performance production systems in Agile environments.

SKILLS

Programming Languages: Java, JavaScript (ES6+), TypeScript, Python, SQL
Frontend: React.js, Angular, Redux Toolkit, Material-UI, D3.js, Bootstrap, Responsive Design, Micro-Frontend Architecture
Backend: Node.js, Express.js, Spring Boot, GraphQL, REST APIs, BFF Architecture, Spring Security, API Gateway, FHIR API Integration
Databases: PostgreSQL, MongoDB, MySQL, SQL Server, Hibernate, JPA, Redis, NoSQL, Data Modeling, Query Optimization, Database Migration
Cloud & DevOps: AWS (EC2, S3, Lambda, ECS, EKS), Azure, Kubernetes, Docker, Terraform, Helm, CI/CD (GitHub Actions, Jenkins)
Testing & Quality: JUnit, Mockito, Jest, Cypress, Integration Testing, Unit Testing, Postman, Performance Optimization, SonarQube
Streaming & Systems: Apache Kafka, Event-Driven Architecture, WebSocket, SSE
Security: OAuth2, JWT, RBAC, Spring Security, Secure Coding Practices
Monitoring: Grafana, AWS CloudWatch, Splunk, Logging & Alerting, Incident Retrospectives
Tools & Methodologies: Git, JIRA, Agile/Scrum, Confluence, Postman, API Documentation, Sprint Planning

EXPERIENCE

CitiusTech Jan 2025 – Present
Full Stack Developer Remote

- Developed a cloud-native healthcare analytics platform serving 10K+ users, leveraging micro-frontend architecture (React) and microservices (Spring Boot, Node.js) to enable modular, independently deployable features.
- Implemented of GraphQL and REST APIs integrated with healthcare/EHR systems, enabling real-time clinical data exchange across distributed services.
- Designed reusable UI component libraries and shared BFF contracts using React.js, TypeScript, and Material-UI, reducing cross-team development time and improving frontend scalability.
- Built and secured RESTful APIs using Spring Security, OAuth 2.0, and JWT, enforcing role-based access control and HIPAA aligned secure coding practices across clinical data applications.
- Automated CI/CD workflows using Jenkins, Docker, and GitHub Actions, cutting release cycle times and enabling zero downtime deployments across environments.
- Improved observability using Grafana and AWS CloudWatch, setting up alerting dashboards across microservices to enable proactive monitoring and faster incident resolution.
- Created unit, integration, and end-to-end test suites using JUnit, Mockito, and Cypress, improving code coverage and reducing production bugs across key application modules.

Infosys Limited Aug 2021 – Jun 2023
Full Stack Developer Hyderabad, India

- Built scalable microservices using Node.js, Express.js, and Spring Boot to handle high-volume vehicle and dealership data workflows, supporting daily API transactions across dealer and customer portals.
- Developed micro-frontend UIs using Angular with lazy loading, code-splitting, and Redux Toolkit state management, improving load times and UX for dealer agents across international markets.

- Designed Backend-for-Frontend (BFF) services in Node.js to aggregate microservices, normalize payloads, and implement caching/pagination, reducing API latency by ~25%.
- Optimized PostgreSQL and MongoDB schemas, indexing strategies, and queries under concurrent load; led a zero-data-loss MongoDB-to-PostgreSQL migration.
- Automated multi-environment deployments using GitHub Actions, Jenkins, Docker, and Kubernetes with Helm-based configurations, reducing manual deployment effort and rollback risks.
- Monitored backend services using AWS CloudWatch, setting up log-based alerts that improved system health visibility and accelerated issue resolution.
- Delivered modules for vehicle verification, digital signatures, and document validation in cross-functional Agile sprints with consistent on-time releases.

Trigyn Technologies Sep 2020 – Jul 2021
Java Full Stack Developer Remote - India

- Built RESTful APIs using Java 11, Spring Boot, and Hibernate to support integration between internal modules in a municipal finance application, reducing manual reconciliation efforts.
- Migrated legacy JSP and Struts-based modules into modular Spring Boot microservices, improving maintainability and cutting deployment time across administrative workflows.
- Developed interactive frontends using React.js, TypeScript, and Bootstrap for government citizen service portals, streamlining case intake and increasing self-service adoption.
- Implemented event-driven processing patterns using Kafka-style messaging for reliable asynchronous data workflows across internal modules.
- Optimized SQL Server stored procedures and queries, reducing report generation time for high-volume datasets.

EDUCATION

BS in Computer Science and Engineering Vijayawada, India
PVP Siddhartha Institute of Technology

MS in Data Science Denton, Texas
University of North Texas

PROJECTS

InkToNote | React, Node.js, TypeScript, PostgreSQL, Prisma, Groq AI, TrOCR, PaddleOCR | GitHub

- Built a full-stack AI-powered handwriting recognition app that converts uploaded images/PDFs into structured digital notes using a hybrid OCR pipeline (PaddleOCR for text detection + TrOCR for handwriting recognition).
- Integrated Groq AI API to intelligently structure raw OCR output into headings, bullet lists, and summaries with per-line confidence scoring.
- Developed a Notion-style block editor (BlockNote) for post-OCR editing, with notebook organization, tagging, full-text search, and Markdown/plain text export.
- Implemented JWT-based authentication, RESTful API with Express/TypeScript, and PostgreSQL schema managed via Prisma ORM.

Patient Appointment API | Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Spring Security, JWT, Docker | GitHub

- Designed and built a RESTful healthcare scheduling API supporting Patient, Doctor, Slot, and Appointment management with role-based access control (ADMIN, DOCTOR, PATIENT).
- Implemented JWT authentication, slot conflict detection to prevent double-booking, and global exception handling with structured JSON error responses.
- Managed database schema evolution using Flyway migrations and containerized the full application with Docker and Docker Compose for local development.
- Documented all endpoints via Springdoc OpenAPI (Swagger UI) for interactive API exploration.